

SolidarityNotCharity

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Summary

Mutual aid is an ancient and theorized biological phenomena observed in many life forms and ecosystems as old as time. The concept is meant to capture the way organisms of the same species care for one another. Theorists of this field claim that evidence shows how mutual support within a species actually makes them thrive in contrast to Darwins' evolutionary theory on survival of the fittest

In modern times the practice of mutual aid attempts to replicate indigenous practices of supporting one another in shared struggle. Mutual aid acknowledges the web of relationships we all live in and how we all lean on each other to survive. Today mutual aid organizations contrast the work of charities for their direct community involvement and acknowledgement that "aid" is not one directional. We are all fulfilled by helping.

This said, mutual aid groups are often operated out of homes, community centers, places of worship, school gymnasiums. Grassroots oriented these groups sometimes operate in silos, have varying access to stable funding, and may have varying presence to be well known by their surrounding community. From a data perspective, there are limited platforms where you can search for several different Mutual Aid Organizations at once. Because mutual aid organizations operate locally, they're often using a multitude of platforms and software to house and organize their data. There are limited platforms where interested parties that would like to donate their time can sign up once and be reached by several organizations. Additionally, donors may struggle to find and connect with local organizations they wish to contribute to. Our service acts as a bridge between mutual aid groups, volunteers, and donors.

There will be different applications for various users. Volunteers can search for mutual aid organizations by filtering based on their locality and social issues of interest. We also offer a feature to be able to send their interest to the mutual aid organization which we achieve through an intake form.

Like the volunteers, mutual aid organizations would also be able to search the database for other organizations which may be of interest to them for partnerships. The benefit for mutual aid organizations to be searchable in our database is their access to our centralized location, the marketing opportunities, and to search for potential donors and volunteers through our database. Donors can more easily find local and small-scale organizations that would benefit from their financial contributions by filtering by issues and states of interest.

Stakeholders

End users/external entities:

- Mutual Aid Organizations
- Volunteers
- Donors

Internal entities

- Development team
- Maintenance and support.

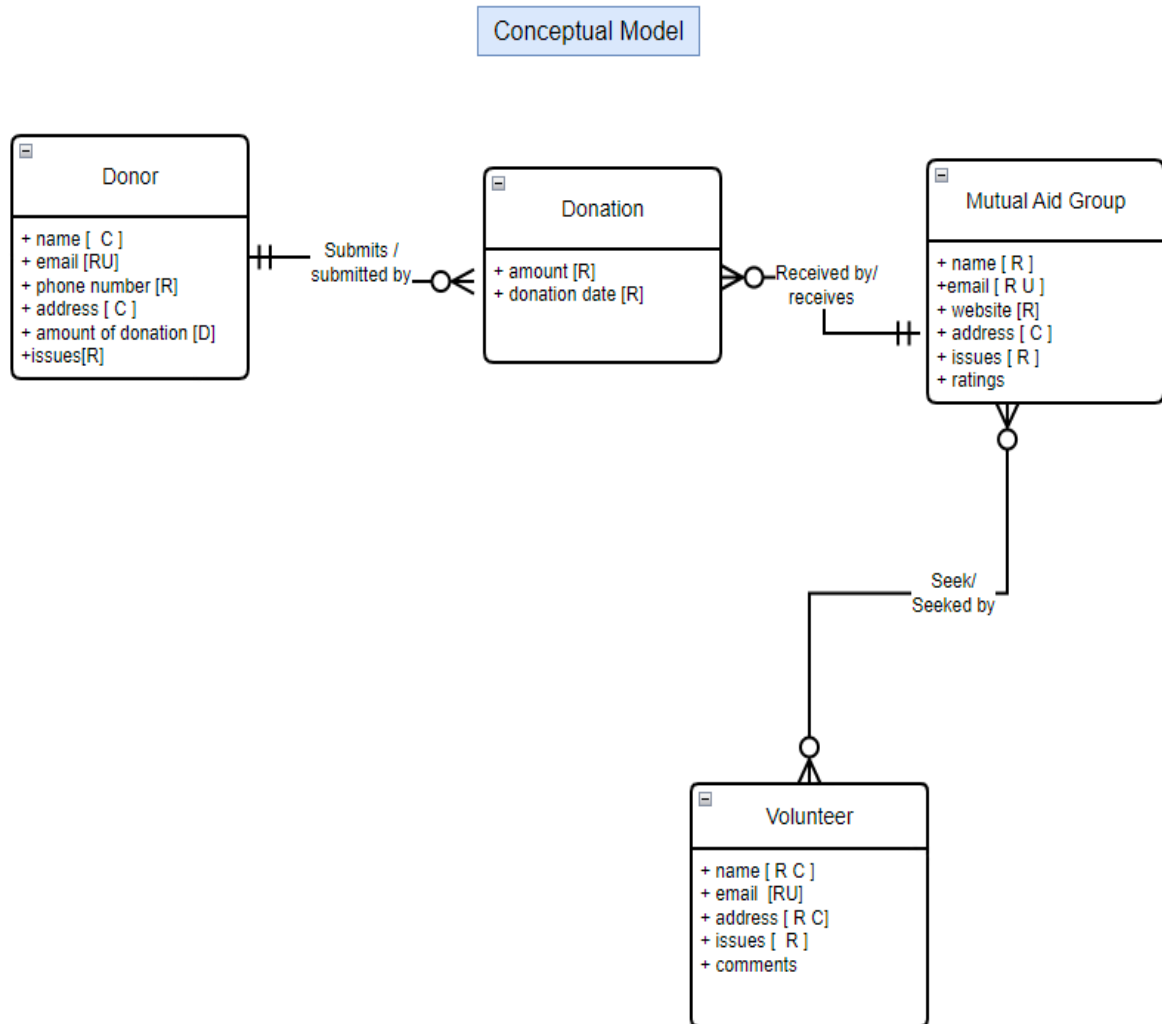
Business rules/assumptions

1. There are limited platforms where you can search for several different Mutual Aid Organizations at once.
2. There are limited platforms where interested parties that would like to donate their time can sign up once and be reached by several organizations.
3. There are limited platforms where interested donors could easily find more grassroots organizations to give their money to.
4. Mutual aid organizations are strapped for resources and do not prioritize marketing due to the nature of their work and limited resources.

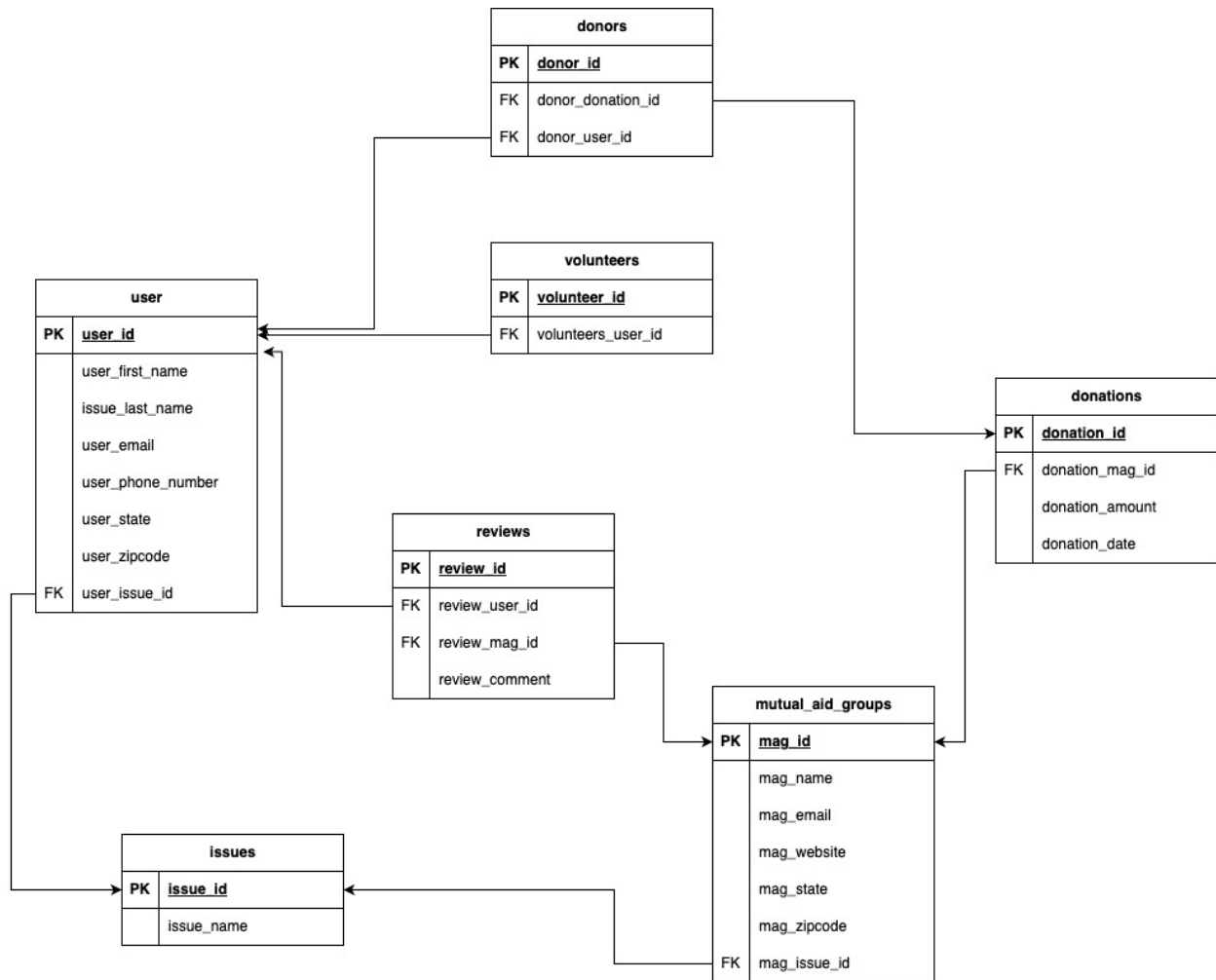
Data Questions

1. Check for volunteers that are interested in a specific issue
2. Check for volunteers with an interest in a specific organization.
3. Check for donors who donated to a specific mutual aid organization.
4. Check for donors with a specific donation dollar amount.

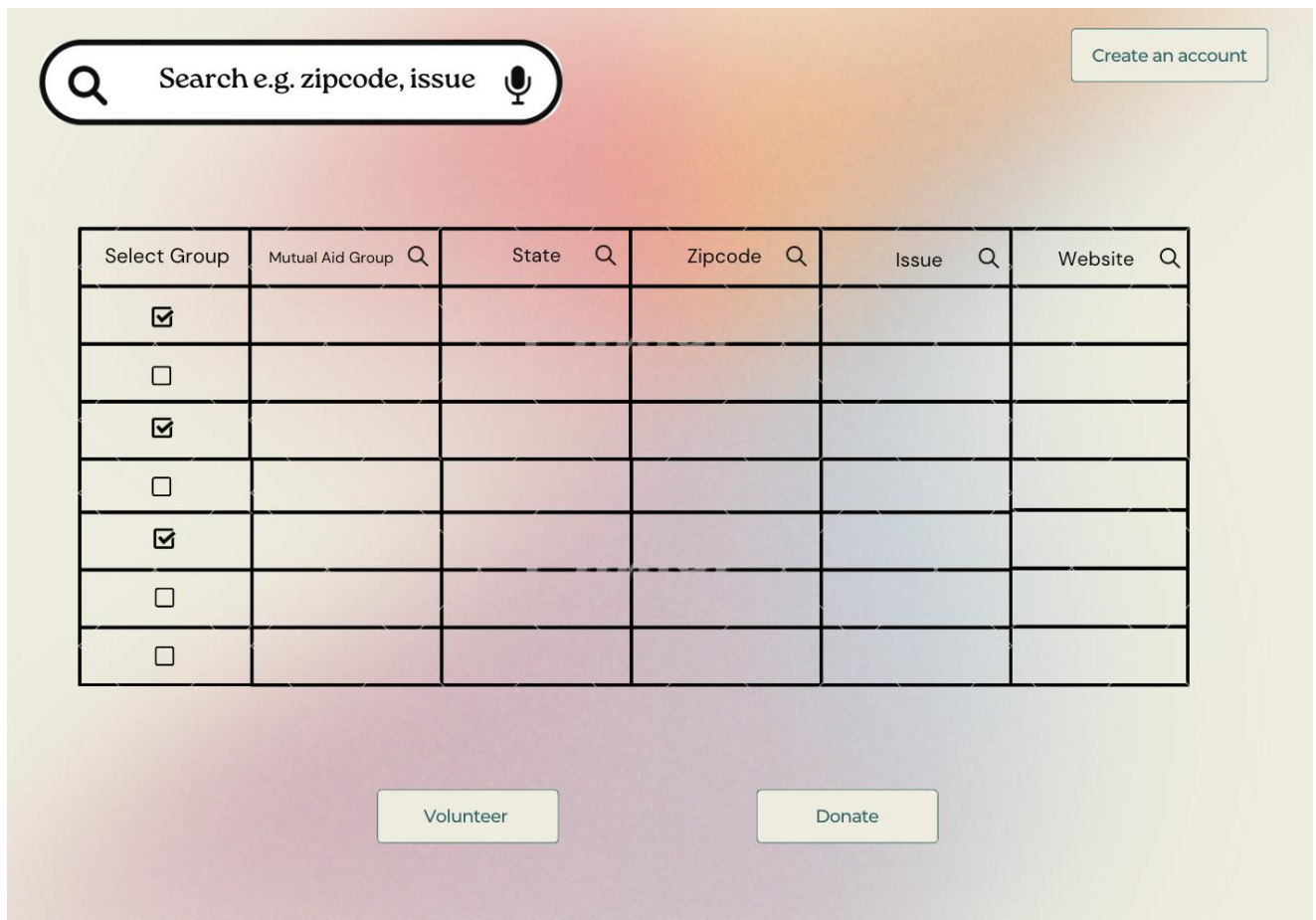
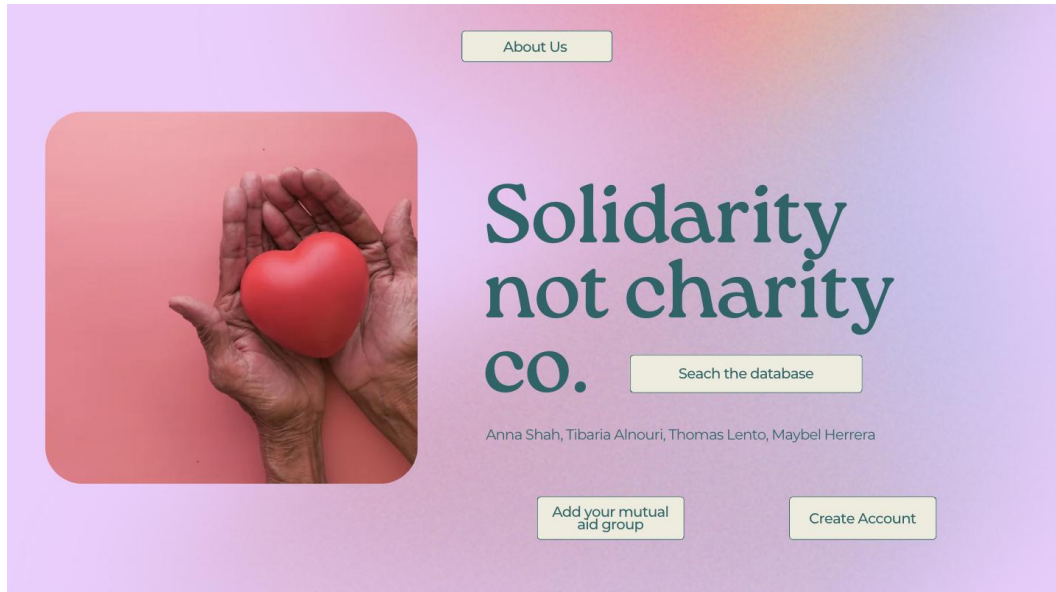
Conceptual Model



Logical Model



Application



Add your mutual aid group to the network!

Submit group

Already Registered? Login

PLEASE ENTER YOUR ORG NAME

e.g. Bike Share Seattle

PLEASE ENTER YOUR EMAIL

PLEASE ENTER YOUR WEBSITE

www.yourwebsite.org or social media

PLEASE ENTER YOUR STATE

e.g. WA

PLEASE ENTER YOUR ZIPCODE

e.g. 98001

PLEASE SELECT AN ISSUE BELOW

e.g. Food Access



Submit group

Sign up to connect with your community!

Create Account

Already Registered? Login

PLEASE ENTER YOUR FIRST NAME

PLEASE ENTER YOUR LAST NAME

PLEASE ENTER YOUR EMAIL

PLEASE ENTER YOUR PHONE NUMBER

PLEASE ENTER YOUR STATE

PLEASE ENTER YOUR ZIPCODE

PLEASE SELECT AN ISSUE BELOW

e.g. Food Access



CREATE A PASSWORD

Create Account

Sign in

Login to Your Account

Not Registered? [Create an Account](#)

PLEASE ENTER YOUR EMAIL

PASSWORD

Login

Send a donation

Donate, ya cheapskate!

MUTUAL AID GROUP DONATING TO:

Bike Share Seattle

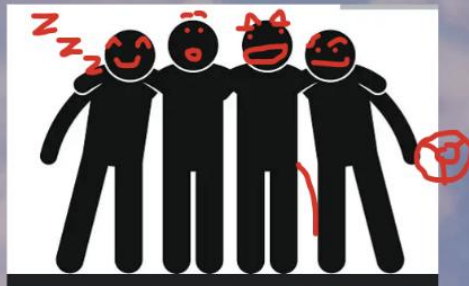
(auto-fills to group selection)

PLEASE ENTER YOUR PAYPAL ACCOUNT

DONATION AMOUNT

Send Donation

about us



Physical Database Design

UP/DOWN SCRIPT ORDER

Down script for softdrops-

- 1.fk_donors_donor_donation_id
- 2.fk_donors_donor_user_id
- 3.Table donors
- 4.fk_reviews_review_user_id
- 5.fk_reviews_review_mag_id
6. Table reviews
- 7.fk_volunteers_volunteer_user_id
- 8.Table volunteers
- 9.fk_users_user_issue_id
- 10.Table Users
- 11.fk_donations_donation_mag_id
- 12.Table donations
- 13.fk_mags_mag_issue_id
14. Table mags
15. Table issues

Up script - creation of tables and foreign keys

- 1.Table issues
- 2.Table mags
- 3.Fk_mags_mag_issue_id
- 4.Table donations
- 5.Fk_donations_donation_mag_id
- 6.Table Users
- 7.Fk_users_user_issue_id
- 8.Table volunteers
- 9.Fk_volunteers_volunteer_user_id
- 10.Table reviews
- 11.Fk_reviews_review_mag_id
- 12.Fk_reviews_review_user_id
- 13.Table donors
- 14.Fk_donors_donor_user_id
- 15.fk_donors_donor_donation_id

```

if not exists(
    select * from sys.databases where name = 'mutual_aid'
)
create database mutual_aid
GO
use mutual_aid
go

```

--Down Script---

```

IF EXISTS (select * from information_schema.table_constraints
    WHERE CONSTRAINT_NAME = ('fk_donors_donor_donation_id'))
ALTER TABLE donors DROP CONSTRAINT fk_donors_donor_donation_id
GO
IF EXISTS (select * from information_schema.table_constraints
    WHERE CONSTRAINT_NAME = ('fk_donors_donor_user_id'))
ALTER TABLE donors DROP CONSTRAINT fk_donors_donor_user_id
GO

```

```

Drop table if exists donors
go

```

```

IF EXISTS (select * from information_schema.table_constraints
    WHERE CONSTRAINT_NAME = ('fk_reviews_review_user_id'))
ALTER TABLE reviews DROP CONSTRAINT fk_reviews_review_user_id
GO
IF EXISTS (select * from information_schema.table_constraints
    WHERE CONSTRAINT_NAME = ('fk_reviews_review_mag_id'))
ALTER TABLE reviews DROP CONSTRAINT fk_reviews_review_mag_id
GO
drop table if exists reviews
GO

```

```

IF EXISTS (select * from information_schema.table_constraints
    WHERE CONSTRAINT_NAME = ('fk_volunteers_volunteer_user_id'))

```

```
)  
ALTER TABLE volunteers DROP CONSTRAINT fk_volunteers_volunteer_user_id  
go
```

```
drop table if exists volunteers  
go
```

```
IF EXISTS (select * from information_schema.table_constraints  
WHERE CONSTRAINT_NAME = ('fk_users_user_issue_id'))  
ALTER TABLE users DROP CONSTRAINT fk_users_user_issue_id  
GO
```

```
Drop table if exists users  
GO
```

```
IF EXISTS (select * from information_schema.table_constraints  
WHERE CONSTRAINT_NAME = ('fk_donations_donation_mag_id'))
```

```
ALTER TABLE donations DROP CONSTRAINT fk_donations_donation_mag_id  
GO
```

```
Drop table if exists donations  
GO
```

```
IF EXISTS (select * from information_schema.table_constraints  
WHERE CONSTRAINT_NAME = ('fk_mags_mag_issue_id'))
```

```
ALTER TABLE mags DROP CONSTRAINT fk_mags_mag_issue_id  
GO
```

```
drop table if exists mags
```

```
go  
drop table if exists issues
```

go

---Up Script--

--Issues table--

```
Create table issues (  
    issue_id int identity not null,  
    issue_name VARCHAR (50) not null  
    Constraint pk_issues_issues_id primary key (issue_id)  
)
```

GO

---Mags table --

```
Create table mags(  
    mag_id int identity not null,  
    mag_name varchar (100) not null,  
    mag_email Varchar (50) not null,  
    mag_website varchar (50) null ,  
    mag_state char (2) not null,  
    mag_zipcode char (5) not null,  
    mag_issue_id int not null,  
    Constraint pk_mags_mag_id primary key (mag_id),  
    Constraint u_mags_mag_email unique (mag_email)  
)
```

GO

---add FK to mags---

```
Alter table mags ADD  
constraint fk_mags_mag_issue_id foreign key (mag_issue_id)  
references issues(issue_id)
```

GO

--Create table donations--

```
Create table donations (
```

```
donation_id int identity not null,  
donation_mag_id int not null,  
donation_amount money not null,  
donation_date date null,  
Constraint pk_donations_donation_id primary key (donation_id),  
)  
GO
```

--Create FK to donations table--

```
Alter table donations ADD  
constraint fk_donations_donation_mag_id foreign key (donation_mag_id)  
references mags(mag_id)
```

GO

---creating the users table ---

```
Create table users (  
    user_id INT identity not null,  
    user_firstname VARCHAR(50) not null,  
    user_lastname VARCHAR (50) not null,  
    user_email VARCHAR (50) not null,  
    user_phone_number CHAR (10) null,  
    user_state CHAR (2) not null,  
    user_zipcode CHAR (5) not null,  
    user_issue_id int not null,  
    Constraint pk_users_user_id primary key (user_id),  
    Constraint u_users_user_email unique (user_email)  
)
```

GO

---Create FK to users table ---

```
Alter table users ADD  
constraint fk_users_user_issue_id foreign key (user_issue_id)  
references issues(issue_id)
```

go

---create table for volunteers--

```
Create table volunteers (  
volunteer_id INT identity not null,  
volunteer_user_id INT not null,  
Constraint pk_volunteers_volunteer_id primary key (volunteer_id)  
)  
GO
```

---Create Foreign key volunteers---

```
ALTER table volunteers ADD  
constraint fk_volunteers_volunteer_user_id foreign key (volunteer_user_id)  
references users(user_id)
```

GO

----Create reviews table--

```
Create table reviews (  
review_id INT identity not null,  
review_user_id INT not null,  
review_mag_id INT not null,  
review_comment text null  
Constraint pk_reviews_review_id primary key (review_id),  
)  
GO
```

---create FK to reviews table ---

```
ALTER TABLE reviews ADD  
Constraint fk_reviews_review_user_id foreign key (review_user_id)  
references users(user_id) ,  
Constraint fk_reviews_review_mag_id foreign key (review_mag_id)  
references mags(mag_id)  
GO
```

---Create donors table --


```
Create table donors (  
donor_id INT IDENTITY not null,  
donor_donation_id int not null,  
donor_user_id int not null,  
Constraint pk_donors_donor_id primary key(donor_id)  
)
```

```
GO
```

```
---FK to donors table---
```

```
ALTER table donors ADD  
Constraint fk_donors_donor_donation_id foreign key (donor_donation_id)  
references donations (donation_id),  
Constraint fk_donors_donor_user_id foreign key (donor_user_id)  
references users(user_id)
```

```
GO
```

```
----Insert issues--
```

```
Insert into issues (issue_name) values  
(  
'food_security'),  
'environmental'),  
'housing_security'),  
'poverty'),  
'indigenous_rights'),  
'water_security'),  
'transportation'),  
'disaster_relief')  
GO
```

---insert users table---

Insert into users (user_firstname, user_lastname, user_email, user_phone_number, user_state,
user_zipcode, user_issue_id)

VALUES

('James', 'Bond', ' james.bond@yahoo.com', 5106429708, 'CA', 94705, 1),

('Lillian', 'Red', 'lillian_red@gmail.com', 4159620708, 'CA', 94108, 2),

('Mary', 'Lou', 'marylou@gmail.com', 2136590809, 'CA', 91356, 4),

('Joseph', 'King', 'jking@yahoo.com', 3107480405, 'CA', 90210, 5),

('Allen', 'Whistle', 'allwhistle21@gmail.com', 7432983728, 'AK', 94517, 6),

('Alice', 'Copper', 'copper4lice@yahoo.com', 8273098764, 'AZ', 85001, 5),

('Bob', 'Odenkirk', 'bodenkirk55@hotmail.com', 6483963109, 'CA', 94101, 6),

('Lester', 'Night', 'l3st3rnight@gmail.com', 1215427539, 'CA', 94101, 7),

('Lorne', 'Malvo', 'malorne1998@gmail.com', 0798658342, 'CA', 94701, 3),

('Molly', ' Silverspoon', 'spoon4molly@gmail.com', 7453108970, 'CA', 90012, 1),

('Jordan', 'Key', 'jkey424@yahoo.com', 2736451983, 'CO', 80014, 2),

('Mika', 'Monrow', 'mmonrow@hotmail.com', 1092856473, 'HI', 96795, 4),

('Jesse', 'White', 'wh1tej3ssie@yahoo.com', 1092184736, 'NY', 11224, 5),

('Maya', 'Patel', 'patel.maya0104@gmail.com', 1029098574, 'OR', 97035, 6)

--insert for mags---

```
Insert into mags (mag_name, mag_email, mag_website, mag_state, mag_zipcode, mag_issue_id)
values
(' Mutual Aid Network of Anchorage', 'anchoragemutualaid@gmail.com', 'wearemana.org', 'AK',
94517,6 ),
('Mutual Aid Phx', 'mutualaidphx@gmail.com', 'www.mutualaidphx.com', 'AZ', 85001,7 ),
('SF Mutual Aid ', 'sfbayareamutualaid@gmail.com', 'www.sfmualaid.com', 'CA', 94101,4 ),
('Berkeley Mutual Aid', ' info@berkeleymutualaid.org', 'www.berkeleymutualaid.org ', 'CA'
,94701,2 ),
('Mutual Aid Los Angeles Network', ' info@mutualaidla.org', 'www.mutualaidla.org', 'CA',
90012,8 ),
('Denver Mutual Aid ', 'rebecca@denvermutualaid.com', 'www.denvermutualaid.com', 'CO',
80014,4 ),
('Hawaii Peoples Fund ', 'peoples@lava.net', 'hawaiipeoplesfund.org', 'HI', 96795,3 ),
('South Brooklyn Mutual Aid', 'southbkmutualaid@gmail.com', 'www.southbkmutualaid.com',
'NY', 11224,2 ),
('Portland Mutual Aid Network', 'portlandmutualaidnetwork@gmail.com',
'www.portlandmutualaidnetwork.com', 'OR', 97035,1)
```

---insert donations---

```
Insert into donations ( donation_mag_id, donation_amount, donation_date)
values
(1, 10.00, '10-23-23'),
(6, 25.00, '9-4-23'),
(3, 50.00, '6-1-20'),
(7, 100.00, '3-14-20'),
(4, 150.00, '2-14-21')
```

GO

-----insert in to donors ---

Insert into donors (donor_donation_id, donor_user_id) values

(3, 1),

(4, 4),

(5, 5),

(2, 8),

(1, 14)

go

---insert in to reviews---

Insert into reviews (review_user_id, review_mag_id, review_comment)

values

(4, 7, 'it is very fulfilling being able to help'),

(6, 7, 'Im fortunate to be able to contribute'),

(3, 2, 'my time and donation was appreciated'),

(4, 8, 'This organization is amazing'),

(2, 1, 'I received help fast and when I needed it the most')

go

---insert in to volunteers---

Insert into volunteers (volunteer_user_id) values

(1),

(2),

(4),

(3),

(5),

(7),

(6),

(9),

(8),

(13),

(12),

(14),

(11),

(10)

--- Data science questions --

-- see volunteers with specific interest in issue --

```

SELECT u.user_firstname, u.user_lastname, u.user_email, i.issue_name
FROM volunteers v
JOIN users u on u.user_id = v.volunteer_user_id
join issues i on i.issue_id= u.user_issue_id
WHERE i.issue_name= 'poverty'
GO

```

Results		Messages		
	user_firstname	user_lastname	user_email	issue_name
1	Mary	Lou	marylou@gmail.com	poverty
2	Mika	Monrow	mmonrow@hotmail.com	poverty

--see volunteers with specific interest in a mag in Hawaii --

```

SELECT u.user_firstname, u.user_lastname, u.user_email, m.mag_name
FROM volunteers v
JOIN users u on u.user_id = v.volunteer_user_id
JOIN mags m on m.mag_issue_id = u.user_issue_id
WHERE m.mag_name = 'Hawaii Peoples Fund'

GO

```

Results		Messages		
	user_firstname	user_lastname	user_email	mag_name
1	Lorne	Malvo	malorne1998@gmail.com	Hawaii Peoples Fund

---See donors who donated to Mutual Aid Phx---

```

SELECT u.user_firstname, u.user_lastname, u.user_email, m.mag_name, o.donation_amount
FROM donors d
JOIN users u on u.user_id = d.donor_user_id
JOIN mags m on m.mag_issue_id = u.user_issue_id
join donations o on o.donation_id= d.donor_donation_id
WHERE m.mag_name= 'Mutual Aid Phx'

```

Results		Messages			
	user_firstname	user_lastname	user_email	mag_name	donation_amount
1	Lester	Night	13st3rnight@gmail.com	Mutual Aid Phx	25.00

--- see donors with donation amount of \$50 or greater--

```
SELECT u.user_firstname, u.user_lastname, u.user_email, o.donation_amount, o.donation_date
FROM donors d
JOIN users u on u.user_id = d.donor_id
JOIN donations o on d.donor_donation_id = o.donation_id
WHERE o.donation_amount >= 50
```

	user_firstname	user_lastname	user_email	donation_amount	donation_date
1	James	Bond	james.bond@yahoo.com	50.00	2020-06-01
2	Lillian	Red	lillian_red@gmail.com	100.00	2020-03-14
3	Mary	Lou	marylou@gmail.com	150.00	2021-02-14

--- see donors with donation amount of \$50 or less--

```
SELECT u.user_firstname, u.user_lastname, u.user_email, o.donation_amount, o.donation_date
FROM donors d
JOIN users u on u.user_id = d.donor_id
JOIN donations o on d.donor_donation_id = o.donation_id
WHERE o.donation_amount <= 50
```

	user_firstname	user_lastname	user_email	donation_amount	donation_date
1	James	Bond	james.bond@yahoo.com	50.00	2020-06-01
2	Joseph	King	jking@yahoo.com	25.00	2023-09-04
3	Allen	Whistle	allwhistle21@gmail.com	10.00	2023-10-23

Reflection

Once the project was completed we realized how important conceptual and logical models are with organizing and writing out the SQL queries in Azure. It was easier to implement the relational model having a visual aid that shows the relationships with each table and how we

needed to connect them with a foreign key. Certain tables needed to be created first so there would be a primary key to attach a foreign key command. We also feel that starting with a conceptual model then building on that to create a logical model is very helpful. You start with a simple layout which gives you a good starting point to see how to break up the tables and where bridge tables need to be inserted.

The most challenging part of this project was making sure that our tables were all connected in some way and that our application met the requirements of our conceptual and logical models. As a team we were able to develop all of these components, bouncing ideas off of each other to end with a result we are confident with. However towards the end of our project there were some things we noticed we could have done differently. For example, how we connect the volunteer and donor tables to the users table. We also debated whether it would have been beneficial to add a states table to eliminate users from inputting invalid characters. Another difficulty we faced was designing the application and how to tie in the donation/ payment methods into our site securely. Despite these challenges, our project was successful in meeting the goals we had placed at the beginning of the quarter.